

imgs2imgs: Improving Visual Consistency in Multiview Image Editing

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Abstract. In this work, we propose `imgs2imgs`, a lightweight training-free pipeline for making visually consistent edits to multiview image sets. Our pipeline operates on an image-to-image basis, applying partial forward diffusion to the original input to preserve its core structure while facilitating localized edits during the guided denoising process. However, unlike standard image-to-image approaches that operate on a single view and often allow global drift from one view to the other, `imgs2imgs` performs localized edits across multiple photos of the same scene. Our method builds on depth-conditioned ControlNet diffusion models and introduces two key innovations. First, we incorporate a mask-guided editing mechanism. By inverting the segmentation mask fed to the DDIM sampler, we force the diffusion process to be confined only to the content within the object-of-interest’s boundaries, while leaving the background unchanged. Second, to maintain cross-view semantic alignment, we introduce a training-free feature sharing mechanism. During the first image passing, the reference, we create a catalog of indexes and values of the corresponding U-Net middle-block features. Then, during the target pass, we inject the feature for each target location into the U-Net’s middle block. Together, these techniques enable high-fidelity, targeted edits across multiview scenes. Achieving the structural preservation benefits of image-to-image partial diffusion, the conditioning guidance of ControlNet signals, and the cross-view alignment necessary for consistent multiview editing through feature sharing, all without requiring retraining or specialized inpainting models.

Keywords: Image editing · multiview image editing · multiview consistency · latent diffusion models.

1 Introduction

Image generation has improved a lot and grew to have its own field of interest in computer vision. The improvement of image synthesis quality through GANs [20] and Diffusion models [11] allowed us to generate high fidelity images that can compensate for images taken by a real camera or animated through a complicated

computer software. This leap in image generation resulted in a further more interest in not only generating the images but also editing them and guiding the changes using natural language.

Models like Pix2Pix [10] and Prompt-to-Prompt [2] have solved a lot of image editing problems using different approaches. These approaches fix part of the challenge, offering control over the fine-grained details in the generated images, and this remains an active and widely researched area. Yet, this work has largely focused on a single shot or a single view. While this has its own use cases and value, it is certainly a constraint in many real-life scenarios. On the other hand, multiview images are a collection of photographs that depict the same physical setting but from different camera poses, sharing overlapping objects and geometry.

In this research, we punctually define multiview images by separating it into two distinct ways, each group having 2 characteristics. The first characteristic is **focus**: the scene could be *object-centric*, *diva* style, where one object is the center of the scene, In contrast, it can be *scene-centric* style where several objects appear in the scene with a busy background. The second characteristic is the **camera shift**. *Narrow-baseline* (or small-shift) where the images are captured with the camera nudged only a few degrees between shots. The other is *wide-baseline* (or large-shift) where the viewpoints are much further apart from each other, sometimes 120° or more, revealing portions of the scene that are completely occluded in the other view. The object-centric and narrow-baseline is common when scanning the object itself to generate a 3D version, editing this category of multiview images is a largely solved problem. Figure 1 shows the multiview image where there is one object appearing in the image and where the camera angle in the first and second view is slightly shifted a few degrees.



(a) First view - Object-centric



(b) Second view - Object-centric

Fig. 1: Shows multiview with Narrow-baseline and small-shift [4]. This object-centric setting is a largely solved problem, and common in 3D scanning

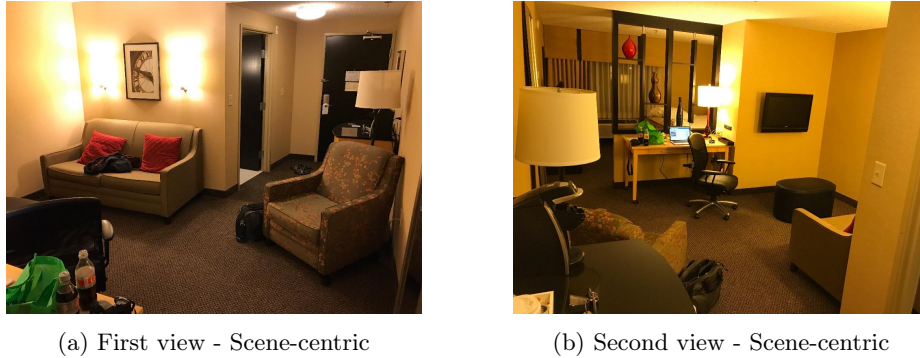


Fig. 2: Shows multiview with Wide-baseline and large-shift. This challenging scene-centric setting is the focus of our research

In contrast, Scene-centric wide baseline is an under researched area of multiview image editing. Figure 2 shows the other category of the images where we might not even realize that those two images are showing the same scene due to high occlusion and large camera angle shift, we can also notice that in these settings there are multiple objects in the scene with a very busy background. Between the two examples we showed, we will be focusing on the second type of image pairs, which are closer to the second category, and this will be the main focus of our research.

Why do we care about multiview image editing? There is no shortage of use cases for how mastering multiview image editing can be beneficial. For starters, 3D reconstruction using techniques like NeRF operates by using multiview shots to create 3D objects of the scene. Having the ability to make one change in one image and generalize it consistently across other views is crucial for VR and creating an immersive digital world. It also has immediate practical value in media production, where a director can simply make one change to a scene and have this change reflected on all other views across footage from every camera. And finally, and perhaps more importantly, solving this problem will allow us to generate incredibly valuable training data for foundational AI models. By making a consistent edit, we can create an aligned data triplet of *Original Images*, *Edit Prompt*, *Edited Images*, and this data is invaluable for image editing and composed image retrieval models.

Yet, *editing wide-baseline scene-centric* multiview images remains a significant challenge. Even for image editing models that relatively succeed in editing single images, they fail if we use them in multiview scenarios. For instance, Figures 3 and 4 show how the InstructPix2Pix model struggles when prompted “Make the painting red” in two different examples of multiview image editing. We can see how the output suffers from poor localization and fails to isolate the specific object of interest and apply the change in color, and instead casts a red tint all over the room.



Fig. 3: InstructPix2Pix results for “Make the painting red”, First View. The edit “bleeds” and incorrectly changes the entire scene colors.

A second, and more subtle, failure mode is stylistic inconsistency. Figure 5 shows the ControlNet [24] model where we tell it to “Make the bed red”. The model might relatively achieve the required edits but fails to generalize them over both views, and we can see that the results for each view are stylistically inconsistent, as the model hallucinates two completely different designs. A noticeable thing about ControlNet is that while it fails to maintain consistency across the two views, it preserves the structure of the original images using conditional signals, like depth maps, segmentation, or Canny edges.

So, we need a method that can solve both localization and consistency problems.

To address these limitations, we propose `imgs2imgs`, A mask-guided editing mechanism that uses partial forward diffusion to ensure precise localization while preserving the original scene’s structure .

A novel, training-free cross-view feature sharing mechanism (using the middle-block feature catalog) that solves the consistency problem.

2 Related Work

Image editing is a long-standing challenge in computer vision. It has been addressed through classic methods, such as Poisson image editing, as well as learning-based approaches. Of course, learning-based methods have prevailed and largely taken over, dominated by generative models like diffusion models



Fig. 4: InstructPix2Pix results for “Make the painting red”, Second View. This demonstrates the same poor localization failure from another viewpoint.



Fig. 5: Prompt: “Make the bed red/green”. This demonstrates poor localization, as the edit incorrectly splashes color onto the walls and doors, not just the bed.

[18][11] and GANs [20][8]. We can broadly categorize the field into single-image editing and 3D-aware editing.

For single-image edits, methods typically fall into two categories. 1) Editing with a fixed backbone, which modifies the diffusion process rather than the model weights. Examples include noise-injection methods like Blended Diffusion [1], DiffEdit, Imagic [6], and UniTune [23]. For instance, MagicMix [13], which begins from a real image rather than pure Gaussian noise. They can use attention-steering methods, Prompt-to-Prompt [10], for example, offers control

by manipulating cross-attention maps during the diffusion process. 2) Retraining or fine-tuning the backbone itself, like Aesthetic Embeddings [21], Textual Inversion [7], and HyperNetworks [9]. To give an example, Pix2Pix [2], which learns a direct mapping from an input image to a target output using paired datasets, DreamBooth [19], which fine-tunes the entire diffusion model. Also, LoRA (Low-Rank Adaptation) [12] adds trainable matrices while freezing the weights. A hybrid approach like ControlNet [24], where they build on the backbone as-is but add a parallel, trainable structure for conditional signals and fine-tune that instead.

As we have mentioned in the introduction, whenever these single-image editing models are used for multiview images, they suffer from two issues: *poor localization* and, most importantly, *inconsistent style*.

Another category is 3D-Aware Editing, which uses 2D diffusion models as a tool, such as DGE [5] and Attention feature [15]. Take for example, MVInpainter [4], which bridges 2D edits to multiview and ensures cross-view consistency using video priors. However, these methods require several views of the object, additional information (i.e., known camera poses and calibration matrices), are heavyweight, and most importantly, target the type of multiview images that we categorized as object-centric, narrow-baseline views. Figure 6 shows one example of the multiview images on which MVInpainter operates.



Fig. 6: An example of object-centric narrow-baseline multiview images [4].

3 The imgs2imgs Method

3.1 Pipeline Overview

Our model `imgs2imgs` addresses the main 2 problems we discussed. Another pitfall many editing models fall into is not preserving the structure of the image. This is why we build on top of the ControlNet model; though it suffers from both poor localization and stylistic inconsistency, Figures 7 and 8, its spatial conditioning signals guide the generation process, preserving the structure of the image. Figure 9 shows the structure of ControlNet, which is built on top of Stable Diffusion with a standard U-Net, an encoder, and a decoder, and a

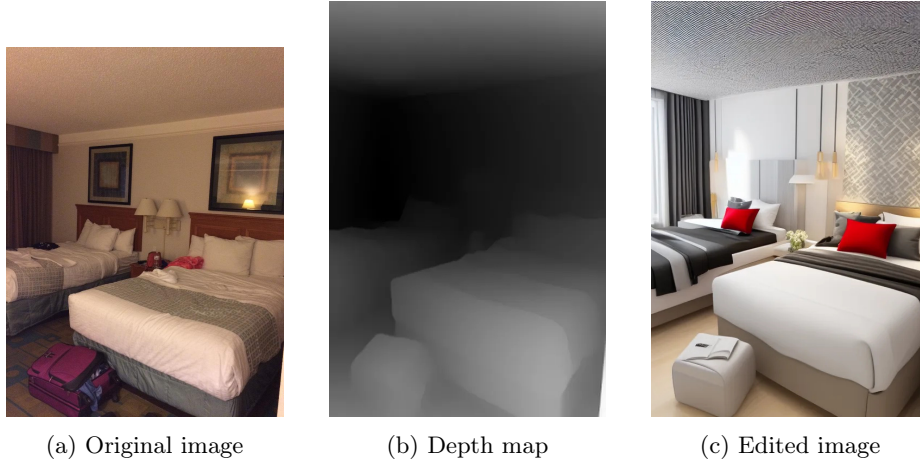


Fig. 7: Editing the first view by the prompt: “make the bed red”. This figure, when compared with Figure 8, demonstrates stylistic inconsistency.

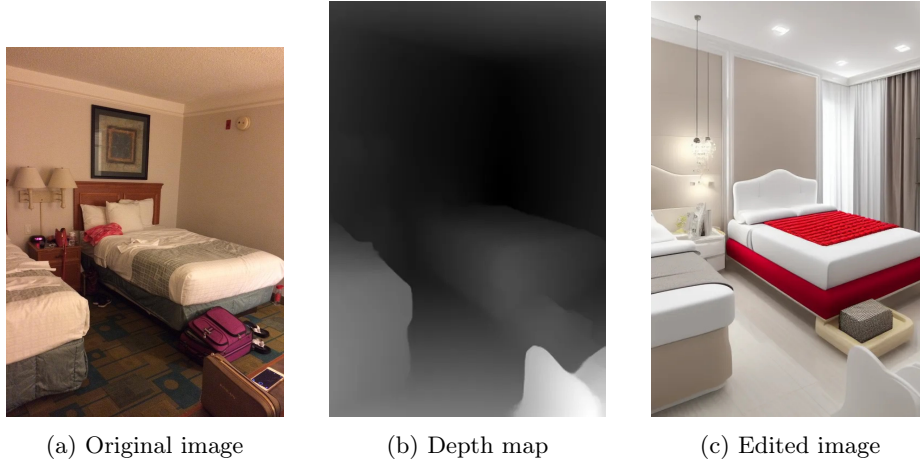


Fig. 8: Editing the second view by the prompt: “make the bed red”. This result is stylistically different from the edit of the first view.

parallel path that feeds in conditioning information—in our case, a depth map Midas model [17].

Our method, `imgs2imgs`, addresses the two key challenges of localization and consistency that we discussed in the introduction. As shown in Figure 10, our pipeline is built on top of a pre-trained, depth-conditioned ControlNet model, which is used for its strong ability to preserve scene structure.

The full process takes four inputs: a reference view, a target view, an editing prompt (e.g., “make the bed runners black”), and a segmentation prompt (e.g.,

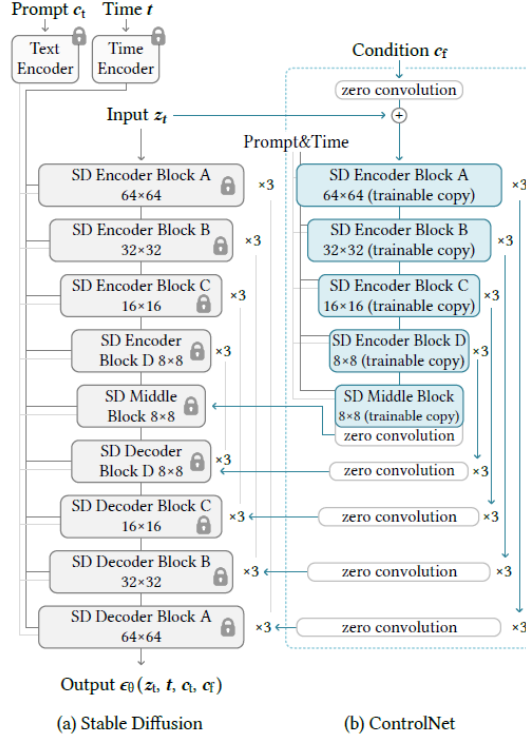


Fig. 9: Stable Diffusion’s U-Net architecture with a ControlNet. Figure from [24].

"bed"). The pipeline then performs two main tasks in parallel. First, it uses the segmentation mask to ensure the edit is localized. Second, it uses our novel cross-view feature sharing mechanism to enforce stylistic consistency. The following subsections will detail each of these two contributions.

3.2 Mask-Guided Localized Editing

To perform a targeted edit, we must ensure the changes are confined only to the object of interest, leaving the background untouched, Figure 11. Instead of starting from pure noise, we apply a partial forward diffusion pass to the original image. This helps preserve the global structure and high details of the scene. We generate a binary mask M on-the-fly from a text phrase using CLIPSeg [14]. This mask is used to composite the latent while keeping the background from the original latent and only use the newly generated content for the pixels inside the mask, as described by **Equation** (1). To avoid harsh edges and ensure a seamless blend, this mask is then feathered using a two-pass `ImageFilter.GaussianBlur` (first with a radius of 3.0, then 10.0). At each denoising step, we use this mask

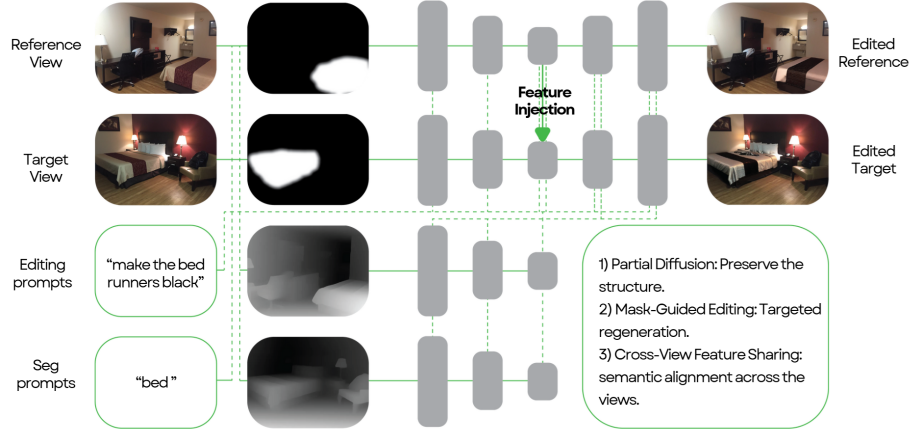


Fig. 10: The **imgs2imgs** pipeline. Our method takes a reference view, a target view, and two text prompts (for editing and segmentation) as input. It uses a depth-conditioned ControlNet backbone. Our key innovations are (1) a mask-guided editing process for localization and (2) a novel **Feature Injection** mechanism between the U-Net middle-blocks to enforce cross-view consistency.

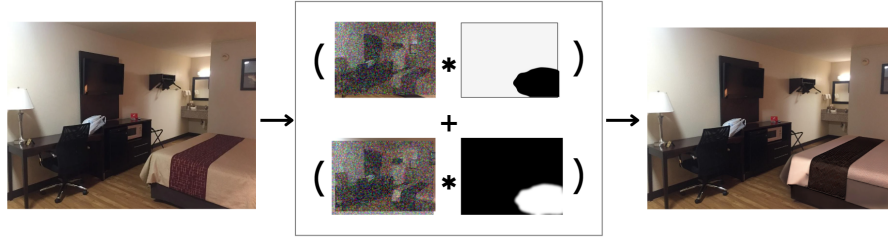


Fig. 11: Partial diffusion and Confining the edits to the target object while keep the rest of the image as is.

to composite the latent. We keep the background from the original latent and only use the newly generated content for the pixels inside the mask.

$$z'_t = z_{t,orig} \odot (1 - M_{obj}) + z_{t,gen} \odot M_{obj} \quad (1)$$

Finally, after the full denoising process is complete and the latent is decoded, we apply a statistical color correction step. This matches the mean color of the generated patch to the mean color of the original image, ensuring the edit blends in with the scene's original lighting and has the same hue color.

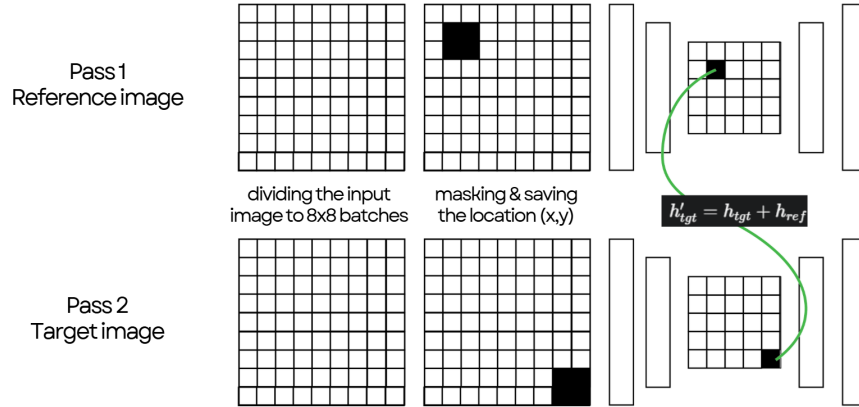


Fig. 12: An overview of our two-pass, training-free Cross-View Feature Sharing mechanism. **(Pass 1 - Reference View)** We process the reference image and create a catalog, storing the U-Net middle-block features from the masked object region, mapped to their spatial indices. **(Pass 2 - Target View)** We process the target image and, at the corresponding U-Net middle-block locations, we inject the saved features from the catalog. This acts as a shared semantic anchor to enforce stylistic consistency.

3.3 Cross-View Feature Sharing

While the mask-guided editing in the previous section solves localization, the main challenge of stylistic inconsistency remains and it is not addressed enough in the image editing literature. To solve this, we introduce a two-pass, training-free feature sharing mechanism, Figure 12. While feature injection itself is not a new concept, we face a different challenge than prior work. In many models that use injection, both views are nearly identical and fall under the narrow-baseline, object-centric category, where the object’s spatial position in the frame is almost always identical. This is not the case for us, as we are dealing with wide-baseline, scene-centric views where an object’s position and scale can vary significantly. To overcome this, we propose two strategies. Our primary method is an efficient index-based method. First, we observe that for an input image, 512x512, its dimension in the U-Net’s middle block becomes an 8x8 grid. In Pass 1, we process the first view, the reference view, and map the input image to an 8x8 grid. We use the object mask M_{obj} to identify which cells (x,y) in this 8x8 grid contain the object of interest, and we mark the cell where the object is. When the input image’s representation reaches the middle block, we leverage this spatial alignment and for each marked cell, we extract and store its corresponding U-Net middle-block feature vector, creating a catalog of features mapped to their spatial (x,y) indices. In Pass 2, we process the second view, the target view. We will map the input image again to an 8x8 grid and mark the cell where the object

is located at. As the U-Net processes the target image, in the middle block and at the exact same cell indices (x,y) that were marked, we inject the saved reference features, adding them to the target view’s features. This process is repeated at each DDIM denoising step, which forces the generation of the object in the target view to be guided by the features from the reference view. This acts as a shared semantic anchor, ensuring the two views converge on a consistent style and design. This index-matching approach works best when the object occupies a roughly similar spatial area in both views. However, in cases with significant differences in scale of the object, we also developed a more robust semantic-matching alternative. In this mode, the catalog in Pass 1 will also store the indices and the values of the object feature, but we will also calculate its CLIP embedding. Then, during Pass 2, we use a Nearest-Neighbor Search to find the most semantically similar feature from the catalog to inject, rather than relying on a simple (x,y) index match. This ensures the correct features are shared even if the object is in a different location in the frame.

4 Experiments and Results

4.1 Experimental Setup

To evaluate our `imgs2imgs` method, we test it against several baseline models mentioned in our related work, which are: `LEDITS` [22], `pix2pix` [2], `Self-Distillation` [3], and a baseline `ControlNet` [24].

To measure performance, we use two categories of metrics. First, to measure image quality and fidelity to the original scene, we use the standard metrics of SSIM and PSNR, where higher values are better. Second, to address the core problem of this paper, we evaluate edit consistency. For this, we propose using CLIP Similarity [16]. We calculate this by taking the two edited output images from a pair, passing them through a pre-trained CLIP image encoder, and calculating the cosine similarity between their resulting feature vectors. A higher score (closer to 1.0) indicates a higher degree of semantic and stylistic consistency between the two edits.

4.2 Qualitative Results

We first show our qualitative results as a visual comparison against the baseline.

The left column shows the original pair of images. The middle column shows the output from the baseline `ControlNet`. As we can see, it fails to produce a consistent edit. When given the same prompt, it hallucinates two completely different styles for the bed, failing the consistency test. The right column shows the result from our model, `imgs2imgs`. Our method successfully generates an edit that is both cleanly localized to the target object and, most importantly, visually consistent across both views.

Figures 17 and 18, and show more qualitative results:

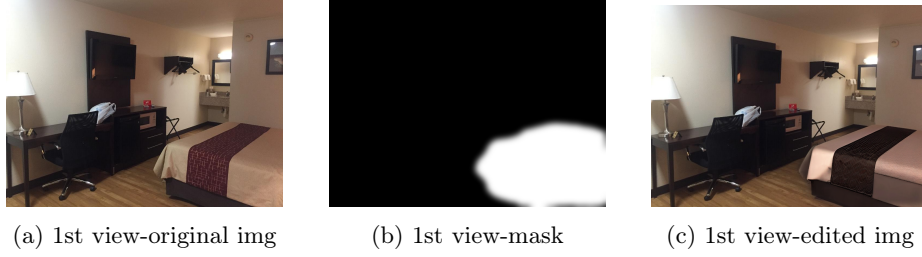


Fig. 13: An illustration of our mask-guided editing process on the first view. From left to right: (a) the original image, (b) the feathered mask generated from the prompt “bed”, and (c) the final edited result. The edit is successfully localized to the target object.

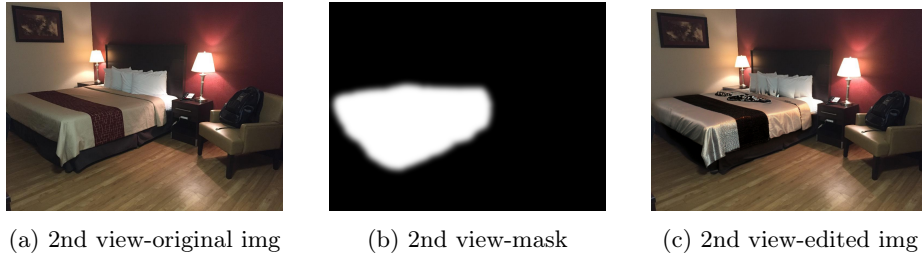


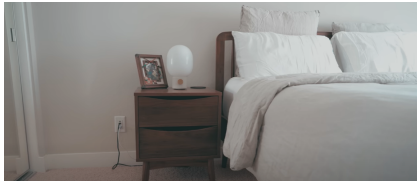
Fig. 14: The same mask-guided process applied to the second view. (a) Original image, (b) generated mask, and (c) the final edited result. This demonstrates that our localization method is effective across different viewpoints.



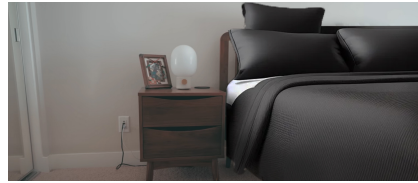
Fig. 15: Qualitative comparison of our method against baseline ControlNet. While ControlNet preserves the scene structure, it lacks consistency, hallucinating two completely different styles for the bed. Our method, *imgs2imgs*, produces an edit that is both high-fidelity and stylistically consistent across both views.



Fig. 16: Qualitative comparison of other baseline models. These methods often suffer from poor localization and global drift. Notice how **Self-Distillation** and **LEDITS** alter the global lighting and style of the entire room, failing to preserve the background.



(a) First view - Original



(b) First view - Edited



(c) Second view - Original



(d) Second view - Edited

Fig. 17: Qualitative result of our method on a wide-baseline, scene-centric pair with the prompt: “make the bed black”. The edit is applied consistently across both views, maintaining the scene’s original structure and lighting.



(a) First view - Original



(b) First view - Edited



(c) Second view - Original



(d) Second view - Edited

Fig. 18: Editing a wide-baseline, scene-centric armchair with the prompt: “make the armchair black”. While the main body of the chair is consistently edited, a limitation is observed in the first view where the wooden legs of the chair are incorrectly changed to white. This is an area for future improvement.

4.3 Quantitative Results

The qualitative results are also backed up by our quantitative metrics. We first evaluated all models for image quality using SSIM and PSNR, where higher is better. Table 1 shows these results. We can see that while methods like LEDITS score very high, our model, **imgs2imgs**, is also a top performer, achieving a leading score in both SSIM and PSNR. This confirms our edits are high-fidelity. However, these metrics just measure image quality, not consistency.

Table 1: Image Quality comparison using SSIM and PSNR. Higher is better. Our model achieves a top-tier score, showing high fidelity.

Model	SSIM $\uparrow$	PSNR (dB) $\uparrow$
LEDITS	0.5643	10.36
imgs2imgs (our model)	0.5503	10.02
pix2pix	0.5025	9.42
Self-Distillation	0.4907	9.12
controlNet	0.3972	6.70

To measure consistency, we used our new CLIP Similarity metric. Table 2 shows these scores. First row is the similarity between the original views with each other as a reference point. A model like **Self-Distillation** gets a very high consistency score, but if you look back at Table 1, it had a much lower image quality score. Our method, **imgs2imgs**, achieves a consistency score that is very competitive with the top performers, while also leading in the quality metrics. This shows that **imgs2imgs** provides the best overall balance of high-quality, high-fidelity edits and strong cross-view consistency.

Table 2: Edit Consistency comparison using CLIP Similarity. Higher is better. Our model shows the best balance of consistency and quality.

Model	CLIP Similarity $\uparrow$
Original Unedited Pair	0.9467
Diffusion Self-Distillation	0.9429
imgs2imgs (our model)	0.9360
LEDITS	0.9299
pix2pix	0.9081
Baseline ControlNet	0.8683

8. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial nets. In: *Advances in Neural Information Processing Systems*. pp. 2672–2680 (2014)
9. Ha, D., Dai, A.M., Le, Q.V.: Hypernetworks. In: *International Conference on Learning Representations (ICLR)* (2017), arXiv:1609.09106
10. Hertz, A., Mokady, R., Tenenbaum, J., Aberman, K., Pritch, Y., Cohen-Or, D.: Prompt-to-prompt image editing with cross attention control. *arXiv preprint arXiv:2208.01626* (2022)
11. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. *Advances in neural information processing systems* **33**, 6840–6851 (2020)
12. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685* (2022)
13. Liew, J.H., Yan, H., Zhou, D., Feng, J.: Magicmix: Semantic mixing with diffusion models. *arXiv preprint arXiv:2210.16056* (2022)
14. Lüddecke, T., Ecker, A.S.: Image segmentation using text and image prompts. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 7086–7096 (2022), https://openaccess.thecvf.com/content/CVPR2022/html/Luddecke_Image_Segmentation_Using_Text_and_Image_Prompts_CVPR_2022_paper.html
15. Patashnik, O., Gal, R., Cohen-Or, D., Zhu, J.Y., De la Torre, F.: Consolidating attention features for multi-view image editing. In: *SIGGRAPH Asia 2024 Conference Papers*. pp. 1–12 (2024)
16. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Meila, M., Zhang, T. (eds.) *Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 139, pp. 8748–8763. PMLR (2021), <https://proceedings.mlr.press/v139/radford21a.html>
17. Ranftl, R., Lasinger, K., Hafner, D., Schindler, K., Koltun, V.: Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *arXiv preprint arXiv:1907.01341* (2019)
18. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 10684–10695 (2022)
19. Ruiz, N., Li, Y., Jampani, V., Pritch, Y., Rubinstein, M., Aberman, K.: Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2023)
20. Salimans, T., Goodfellow, I., Zaremba, W., Cheung, V., Radford, A., Chen, X.: Improved techniques for training gans. *Advances in neural information processing systems* **29** (2016)
21. Schuhmann, C., Beaumont, R.: Laion-aesthetics: Aesthetic score predictors and subsets of laion-5b. *LAION Blog* (2022), <https://laion.ai/blog/laion-aesthetics/>
22. Tsaban, L., Passos, A.: Ledits: Real image editing with ddpm inversion and semantic guidance. *arXiv preprint arXiv:2307.00522* (2023)
23. Valevski, D., Patashnik, O., Lang, O., Cohen-Or, D., Lischinski, D.: Unitune: Text-driven image editing by fine-tuning an image generation model on a single image. *ACM Transactions on Graphics (SIGGRAPH)* **42**(4) (2023). <https://doi.org/10.1145/3592451>

24. Zhang, L., Rao, A., Agrawala, M.: Adding conditional control to text-to-image diffusion models. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 3836–3847 (2023)